

4. Mechanical Design

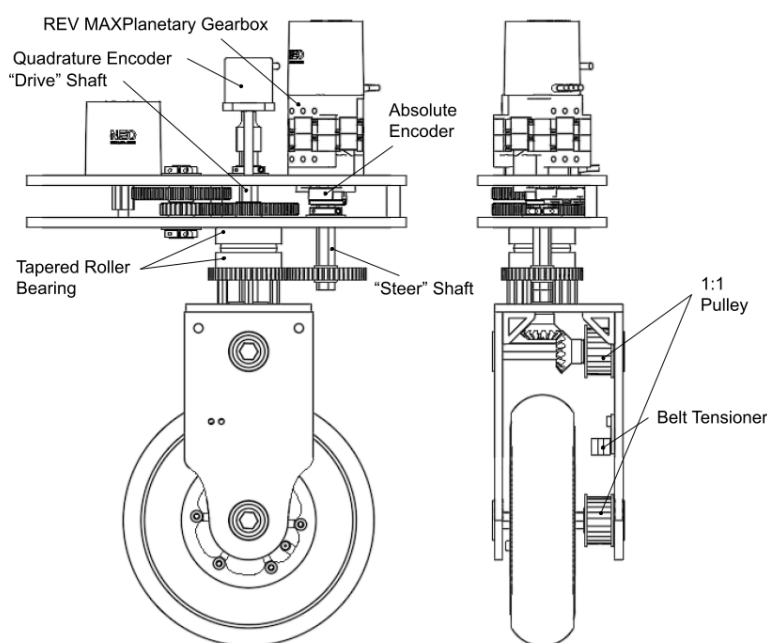
4.1 Overview

The mechanical design of *Twistopher* is focused on making a robot that allows for complete freedom of movement, easy access to electrical subsystems, a complete view of the course, and weather proofing. *Twistopher* has complete control over its direction and position due to the custom-machined swerve modules, and can strafe in any direction, rotate, or strafe and rotate. The electrical systems are easily accessed and wired together thanks to the large, open internals of the chassis with large doors on the top. The tall sensor mast allows the cameras to have a complete view of the course and obtain a highly accurate position of the robot relative to the lanes and obstacles. The robot protects against the elements using acrylic panelling and weatherstripping.

4.2 Swerve Modules

Twistopher features four custom-designed swerve modules that are placed in a square configuration on the robot to form the swerve drive base. The modules are made out of $\frac{1}{4}$ in. machined aluminum. 3D printed prototypes were made to test the assembly of the modules, and also for mounting on the robot for software testing while the metal modules were being machined. Each swerve module is capable of turning the wheel 360-degree continuously while providing full driving force.

The modules have a 15:1 drive gear ratio for a designed top speed of ~8 MPH so that the peak of the motor power curve is at typical IGVC speeds (motors perform optimally at around 60% of their maximum speed). An absolute encoder is installed on the steering shaft and is used to read and maintain the angle of the module between power cycles. Each swerve module features an 8-inch pneumatic wheel, which is ideal for all-terrain and all-weather performance. The modules are designed such that the gearbox is attached to the frame of the robot, so that the modules can be easily removed and installed independently of the gearboxes. This allows for easier assembly and transport, and we can have a spare module fully assembled on standby in case any need to be quickly replaced.



4.3 Frame and Sensor Mast

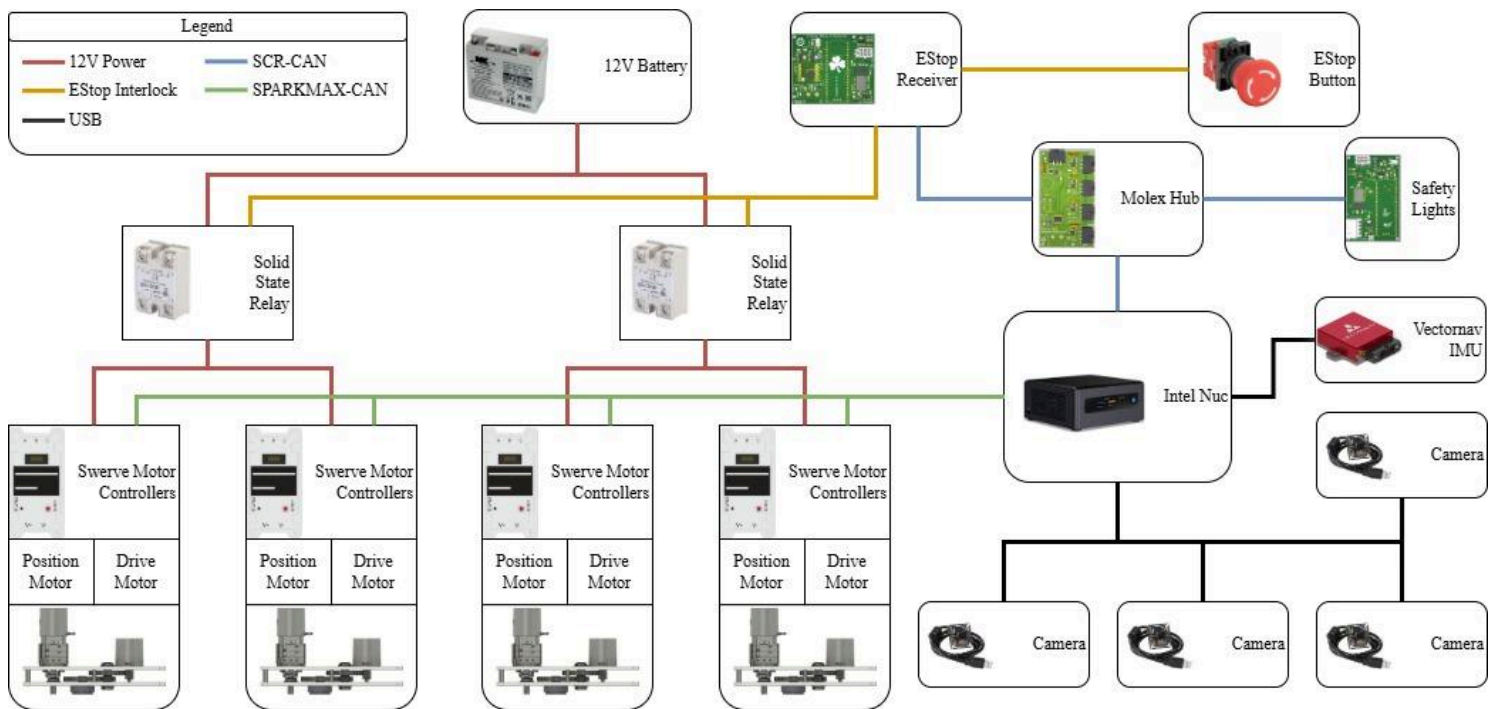
The frame was designed to maximize the performance of the swerve modules. A deliberate effort was made to mount the modules such that the wheels formed a square, as it makes turning in place easier. The frame was also designed for ease-of-access to make tasks like swapping the battery and toggling power easier. Compared to last year's vehicle, *Twistopher* is both wider and taller.

The sensor mast in the middle of the robot was designed to provide a 360-degree view of the course, as the robot can drive in any direction. One of the most important lessons learned from simulating robot parameters is that maximizing the height of the cameras can significantly improve performance. Higher cameras allow for a wider view of the course while also allowing the image to be 2D projected with minimal distortion. Additionally, the GPS antenna, the safety lights, and the emergency stop button are placed on the mast to provide clear visibility and easy access.

4.4 Weatherproofing

Twistopher features large acrylic panels for protection from both rain and sun. Weatherstripping is used to provide a watertight seal around the acrylic. This year, all electronic components are fully enclosed within the acrylic paneling, eliminating the need to route wires through a gasket. The design of the sensor mast, with custom 3D-printed enclosures for each camera, keeps rain off the sensitive electronic components.

5. Electronic and Power Design



5.1 Overview

For 2025, the SCR team continues to use a highly modular approach to electronics to keep repairability and reliability high. *Twistopher's* electronic suite is diagrammed above and features several custom PCBs designed by the team. Every individual electronic module is connected to our custom hub in a star-configuration. The connections are made using 6-pin Molex cables that carry our 12-volt power, data over CAN bus, and the e-stop signal.

By using almost all custom electronics, we are able to implement a custom protocol designed by the team called CONBus, which allows us to remotely configure any device on the vehicle over any bus. For

Twistopher, CONBus is implemented using a specific range of message IDs as part of our CAN bus specification. Using CONBus, the software can quickly modify firmware parameters on any board, which allows for rapid testing.

5.2 Power Distribution

The vehicle uses a single 12V 18Ah AGM lead-acid battery to power both the electronics and the drivebase. Lead acid was chosen for safety, cost, and availability. The high capacity of 18Ah allows for runtimes that far exceed the 5 minutes of runtime needed for the course. Because only a single battery is used, the team can quickly swap out the battery with a fresh one any time the vehicle is operated to make sure the robot is always operating at its peak.

5.3 Motor Control

Twistopher controls the eight motors that make up the swerve drive by commanding the SPARKMAX motor controllers on a separate CAN bus from the main CAN bus. Using separate buses allows us to maximize the throughput of motor control instructions. To drive the swerve drive, four of the SPARKMAX controllers are placed in “position control” which use a high-rate internal PID control loop to set the steering position of each module based on absolute encoder feedback. The remaining four controllers are placed in “velocity control” which allows us to set the desired drive velocity of each module based on quadrature encoder feedback. The main computer can then set the desired velocity state of the robot by computing the desired velocity state of each module and commanding the motor controllers over CAN. By offloading the PID loop onto the motor controllers, we are able to simplify our electronics and further minimize the latency of the robot.

5.4 Safety Lights

Having safety lights on the vehicle to indicate autonomous state is not only required by the rules, but also a feature that we highly depend on for safety. Our safety lights are controlled by a custom PCB, which is capable of reading both the vehicle-wide CAN bus as well as the e-stop line directly. This allows the PCB to immediately update the state of the light when the robot is emergency stopped, while also giving us software control of the light to enable flashing when the robot is enabled in autonomous control.

We chose an RGB LED strip for the safety lights indicator. This is so we can change the color of the lights to use as a debugging tool for evaluating vehicle state while testing and during runs. The color and animation of this strip can be controlled via CAN bus and is implemented in our software to change whenever the robot changes state. The following table shows the map of light color to their meaning.

Light Color	Meaning
White	Standard operation
Yellow	Waypoint following activated
Green	Waypoint reached
Red	Unable to move forward, reversing if possible